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# QMST5336: Course Overview

## About you:



Please complete the “Introduce ME” on *Canvas:Discussion*

- who you are
- your strengths & areas to improve
- your expectations

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# QMST5336: Course Overview

## Classes:

- Thursdays 2:05pm-4:55pm OR 6:35pm-9:20pm
- Synchronous lectures & discussions online

**Office Hours:** Wednesdays 4-6pm & Thursdays 5-6pm

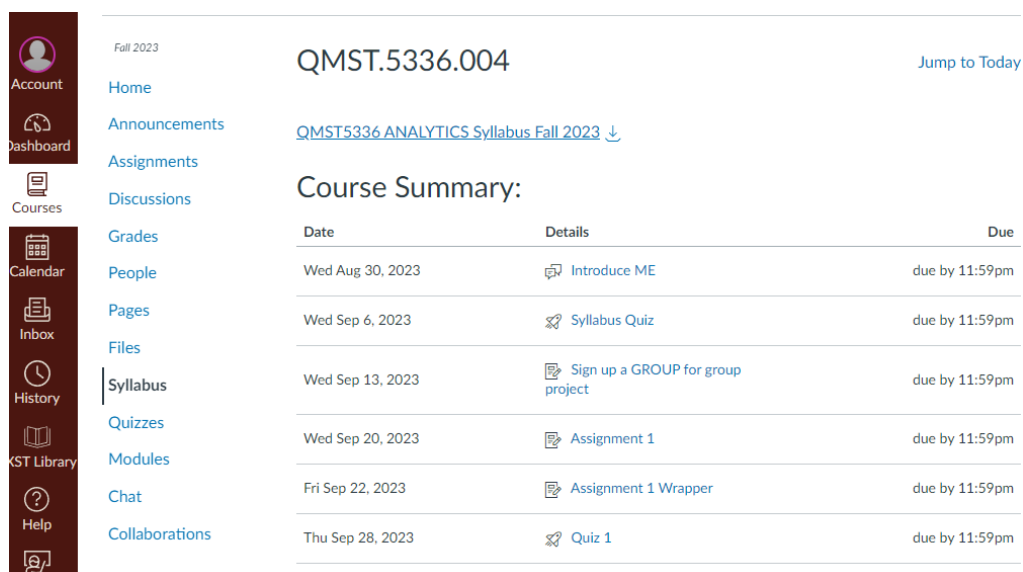
- 15-minute private sessions reserved on Canvas;
- You are allowed to book up multiple sessions for longer meetings;
- First come first serve;
- Join via Zoom at your reserved times;
- *Office Hours are not compulsory.* You can ask questions about the course materials, exams, group project etc.

# DOs in Zoom Meetings

- Please find the passcodes of lectures and office hours on *Canvas:Announcement*.
- Please use your TXSTATE email to log in Zoom.
- Please keep your video/camera on during lectures.
- Please mute yourself if you are not talking.
- Raise hand or type in the chat when you have questions during lectures.
- No discussions during lectures or office hours will be recorded.

## QMST5336: Course Overview

### About the course:



The image shows a screenshot of the Canvas LMS interface for the course QMST.5336.004. On the left is a dark red sidebar with icons for Account, Dashboard, Courses, Calendar, Inbox, History, XST Library, Help, and a video icon. The main content area has a top navigation bar with 'Fall 2023', 'Home', 'Announcements', 'Assignments', 'Discussions', 'Grades', 'People', 'Pages', 'Files', 'Syllabus', 'Quizzes', 'Modules', 'Chat', and 'Collaborations'. The course title 'QMST.5336.004' is displayed in large blue text, with a 'Jump to Today' link to the right. Below the title is a link to the 'QMST5336 ANALYTICS Syllabus Fall 2023'. A 'Course Summary:' section follows, containing a table with columns for Date, Details, and Due.

| Date             | Details                           | Due            |
|------------------|-----------------------------------|----------------|
| Wed Aug 30, 2023 | Introduce ME                      | due by 11:59pm |
| Wed Sep 6, 2023  | Syllabus Quiz                     | due by 11:59pm |
| Wed Sep 13, 2023 | Sign up a GROUP for group project | due by 11:59pm |
| Wed Sep 20, 2023 | Assignment 1                      | due by 11:59pm |
| Fri Sep 22, 2023 | Assignment 1 Wrapper              | due by 11:59pm |
| Thu Sep 28, 2023 | Quiz 1                            | due by 11:59pm |

# QMST5336: Course Overview

## Course Outline:

|                                                                                                                                                                                                                                                                                                                                                                 |                                                                                            |                 |                   |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|-----------------|-------------------|
| McCoy College of Business<br>TEXAS STATE UNIVERSITY<br>Department of Information System & Analytics<br>QMST 5336<br>Analytics<br>Fall 2023                                                                                                                                                                                                                      |                                                                                            |                 |                   |
| INSTRUCTOR:                                                                                                                                                                                                                                                                                                                                                     | Dr. Emily Zhu                                                                              | TEL:            | 512-245-2014      |
| OFFICE:                                                                                                                                                                                                                                                                                                                                                         | McCoy 411                                                                                  | E-MAIL:         | c_z88@txstate.edu |
| OFF. HOURS:                                                                                                                                                                                                                                                                                                                                                     | 4-6pm Wednesdays, 5-6pm Thursdays. Join Zoom meeting ID 820 3429 1979<br>Or by appointment |                 |                   |
| PREFER being contacted by Canvas Inbox messages regarding this course                                                                                                                                                                                                                                                                                           |                                                                                            |                 |                   |
| LECTURE SCHEDULE                                                                                                                                                                                                                                                                                                                                                |                                                                                            |                 |                   |
| Section                                                                                                                                                                                                                                                                                                                                                         | Time                                                                                       | Room            | Zoom Meeting ID   |
| 004                                                                                                                                                                                                                                                                                                                                                             | Th 2:00-5:00PM                                                                             | Encino Hall 204 |                   |
| 005                                                                                                                                                                                                                                                                                                                                                             | Th 6:30-9:30PM                                                                             | McCoy 333       | 853 5961 4672     |
| COURSE DESCRIPTION:                                                                                                                                                                                                                                                                                                                                             |                                                                                            |                 |                   |
| This course introduces analytics, which generally refers to the process of transforming data into information for making decisions. The topics include introduction to analytics, visualization, analytics applications, and challenges related to business data. Students will learn how to use software, conduct data analysis and communicate their results. |                                                                                            |                 |                   |

- In this document you can find everything about the course

# QMST5336: Course Overview

## Course Schedule: (This document will be updated)

| Week                   | Date    | Topics                                                                                            | Milestones                                                                           |
|------------------------|---------|---------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| 1                      | Aug. 24 | Introduce analytics, Big Data, data type and random variables ([T2] §1.3, 1.4, 1.6, 2.2, 4.4)     | Self-introduction due on Aug. 30.<br>Install Python                                  |
| Descriptive Analytics  |         |                                                                                                   |                                                                                      |
| 2                      | Aug. 31 | Python Basics ([T1] §1.4); Visualize quantitative variables ([T2] §2.5 - 2.8)                     | Syllabus quiz due on Sep. 6;                                                         |
| 3                      | Sep. 7  | Categorical variables, missing data, outliers ([T2] §2.9); preprocess data ([T2] §3.1, 3.2, 3.3)  | Sign up groups due on Sep. 13; Group project: visualize data                         |
| 4                      | Sep. 14 | Text mining ([T2] §5.3); Vectorization                                                            | Assignment 1 due on Sep. 20                                                          |
| Predictive Analytics   |         |                                                                                                   |                                                                                      |
| 5                      | Sep. 21 | Clustering ([T2] §5.1); Introduce machine learning                                                | Attendance required for group meeting in class. Assignment 1 wrapper due on Sep. 22. |
| 6                      | Sep. 28 | Linear regression ([T2] §7.4, 7.6, 7.8, 7.10)                                                     | Quiz 1: Sep. 22 – 28                                                                 |
| 7                      | Oct. 5  | Time series ([T2] §8.1-8.5)                                                                       | Assignment 2 due on Oct. 11                                                          |
| 8                      | Oct. 12 | Review; Group Project checkpoint I                                                                | Attendance required for group meeting in class. Assignment 2 wrapper due on Oct. 13  |
| Prescriptive Analytics |         |                                                                                                   |                                                                                      |
| 9                      | Oct. 19 | Formulate optimization problems ([T2] §12.1, 13.1, 13.4)                                          | Quiz 2: Oct. 13 – 19;                                                                |
| 10                     | Oct. 26 | Optimization in Python ([T2] §12.3, 13.3)                                                         |                                                                                      |
| 11                     | Nov. 2  | Simulation ([T2] §11.1); utility ([T2] §15.6); payoff tables and decision trees ([T2] §15.1, 4.4) |                                                                                      |
| 12                     | Nov. 9  | Conditional probability ([T2] §4.3); value of information ([T2] §15.5)                            | Assignment 3 due on Nov. 15                                                          |
| 13                     | Nov. 16 | Final review; Group Project checkpoint II                                                         | Attendance required for group meeting in class. Assignment 3 wrapper due on Nov. 17  |
| 14                     |         | Thanksgiving *****NO CLASS*****                                                                   |                                                                                      |
| 15                     | Nov. 30 | Presentation of Group Projects                                                                    | Quiz 3: Dec. 1 – 7                                                                   |
|                        |         | **Final Exam Period                                                                               | Project Report due on Dec. 8; peer review due on Dec. 9                              |

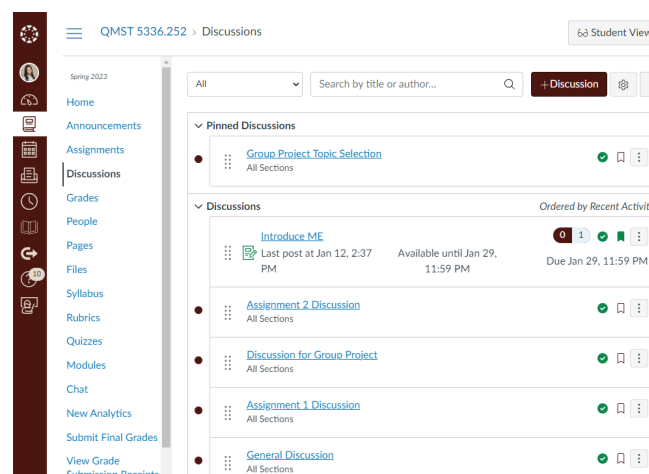
# QMST5336: Course Material

- Your lecture slides (**required**)
- User guide of Python libraries (**strongly recommended**)
  - pandas  
<https://pandas.pydata.org/pandas-docs/stable/>
  - Numpy & SciPy <https://docs.scipy.org/doc/>
  - sklearn  
[https://scikit-learn.org/stable/user\\_guide.html](https://scikit-learn.org/stable/user_guide.html)
- Your course textbook (**Optional but strongly recommended**):  
Camm, D. J., J.J. Cochran et al. (2019). Business Analytics 3rd edition. Cengage Learning: Boston MA.  
Downey, A. (2015). Think Python: How to think like a computer scientist 2nd edition. Green Tea Press: Needham, MA. (Available online:  
<https://greenteapress.com/thinkpython2/thinkpython2.pdf>)



# QMST5336: Course Material

## Questions about the course material:



- Ask questions in **Canvas:Discussions**: It will be much faster than sending messages or emails
- I will try to respond on **Canvas:Discussions** within 24 business hours
- Please feel free to create new discussions if you do NOT find the relevant topics
- **Canvas:Inbox** messages are strongly recommended for personal matters only

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## QMST5336: Grades

### Grading:

- 30% Quizzes (3 quizzes, 10% each)
- 9% Assignments (3 assignments, 3% each)
- 9% Assignment wrappers (3 wrap-ups with group discussion, 3% each)
- 40% Group project
  - 10% presentation
  - 20% report
  - 10% peer review
- 4% Self-introduction
- 4% Group signup
- 4% Syllabus quiz
- 5% bonus for project innovation

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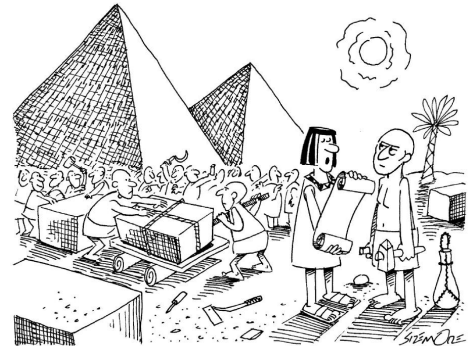
# Teamwork

## Assignments

- You are welcome to discuss assignments with your team members and/or classmates.
- You should submit assignments **INDIVIDUALLY**

## Group Project

- Form a group of 3 or 4 individuals.
- You should team up with others to help your *areas to improve*.
- Team discussion on assignments during designated lectures *via Zoom*.
- Group members help each other throughout the semester.



"You do as you're told, we pay as we please. You work like a slave, we punish at random. That, in a nutshell, is our corporate culture."

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# Teamwork

## Share files

- Canvas group page
- One drive
- Microsoft Teams
- etc.

## Talk & Messages

- Text message
- Zoom
- Microsoft Teams
- What's up
- etc.

# QMST5336: Goals

The goals of QMST5336 are to:

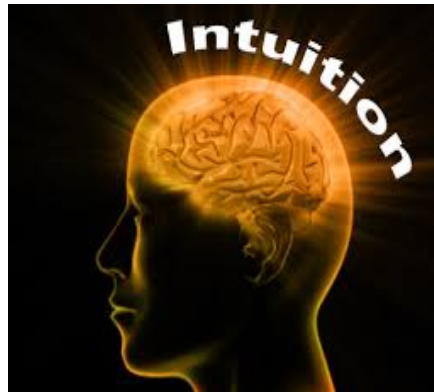
- **Goal 1:** Learn to use data to support decision-making
- **Goal 2:** Be able to use Python for analytics
- **Goal 3:** Learn to interpret analytics approaches and communicate solutions for real-life problems

## Major Drivers of Analytics

- Technological advances, e.g. point-of-sale scanner, IoT (Internet of Things), e-commerce
- Methodological development from ongoing research, e.g. ML (Machine Learning) & AI (Artificial Intelligence)
- Computing power & storage capability, including cloud computing, parallel computing etc.



# Can we trust these approaches to make decisions?



## Challenges in Decision Making

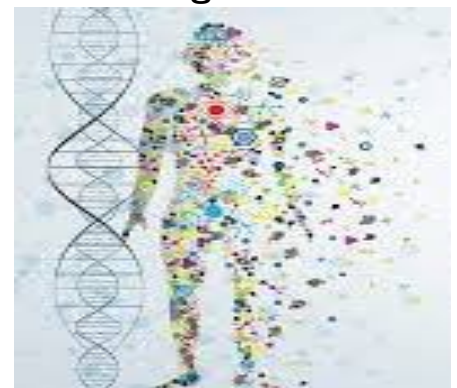
### Uncertainty



### Enormous Amount of Alternatives



### Big Data



# Analytics: The Solution

## Definition

**Business analytics** is the scientific process of transforming data into insight for making better decisions.<sup>1</sup> It is used for data-driven or fact-based decision making, which is often seen as more objective than other alternatives for decision making.

In this course, we use *business analytics* and *analytics* synonymously.

Analytics can help decision making by

- creating insights from data,
- improving our ability to more accurately forecast for planning,
- helping to quantify risk,
- yielding better alternatives through analysis and optimization.

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<sup>1</sup>We adopt the definition of analytics developed by the Institute for Operations Research and the Management Sciences (INFORMS).

## Categorization of Analytical Methods and Models



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# Categorization of Analytical Methods and Models

**Descriptive Analytics:** the set of techniques that describes what has happened in the past.

- data query (a request for information with certain characteristics from a database)
- descriptive statistics
- data visualization including data dashboards
- data-mining (the use of analytical techniques for better understanding patterns and relationships that exist in large data sets)

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# Categorization of Analytical Methods and Models

**Predictive Analytics:** techniques that use models constructed from past data to predict the future or ascertain the impact of one variable on another.

- regression
- time series analysis
- some data-mining techniques
- simulation/risk analysis (the use of probability and statistics to construct a computer model to study the impact of uncertainty on a decision)

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# Categorization of Analytical Methods and Models

**Prescriptive Analytics** finds a best course of action to take; that is, the output of a prescriptive model is a best decision.

- optimization models (models that give the best decision subject to the constraints of the situation)
- simulation optimization (combines the use of probability and statistics to model uncertainty with optimization techniques to find good decision in highly complex and highly uncertain settings)
- decision analysis (used to develop an optimal strategy when a decision maker is faced with several decision alternatives and an uncertain set of future events)
- utility theory (which assigns values to outcomes based on the decision maker's attitude toward risk, loss and other factors)

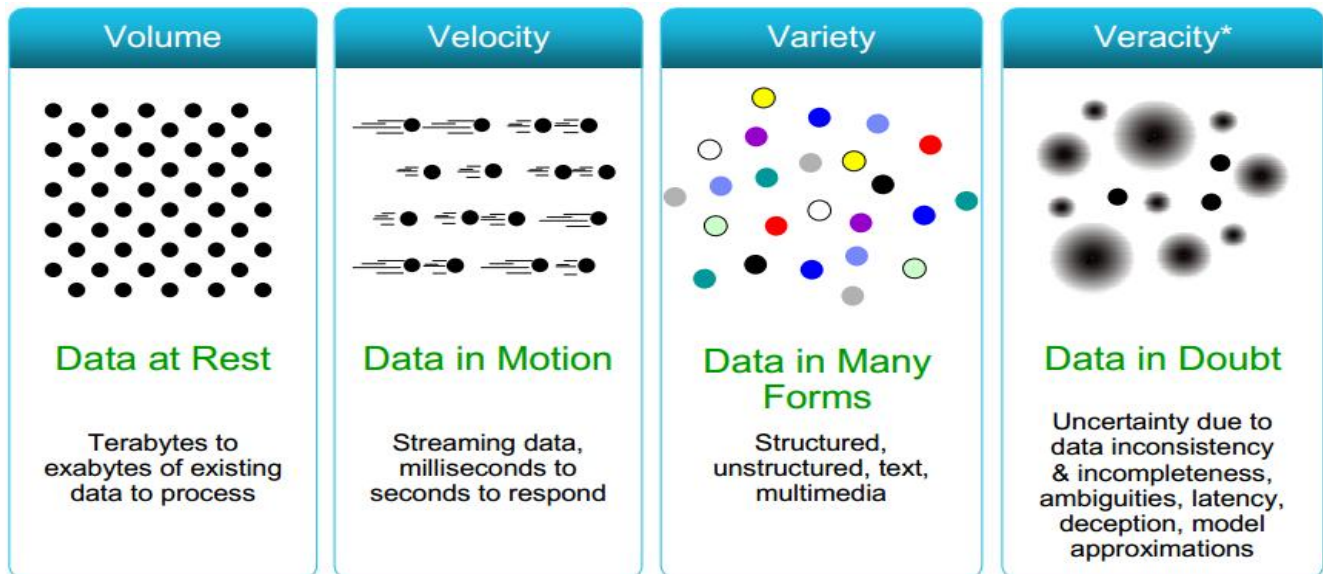
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## This Course

| Topic              | Descriptive | Predictive | Prescriptive |
|--------------------|-------------|------------|--------------|
| Data Visualization | ✓           |            |              |
| Outlier Detection  | ✓           |            |              |
| Missing Data       | ✓           |            |              |
| Clustering         |             | ✓          |              |
| Regression         |             | ✓          |              |
| Time series        |             | ✓          |              |
| Optimization       |             |            | ✓            |
| Decision Analysis  |             |            | ✓            |
| Utility Theory     |             |            | ✓            |
| Simulation         |             |            | ✓            |

# Big Data

**Big data** is any set of data that is too large or too complex to be handled by standard data-processing technics and typical desktop software.



Source: IBM

## The 5th V of Big Data

### The **Value** of Big Data in Service-Oriented Organizations

Over 75% of organizations in professional services; engineering, construction, and operations; cargo transportation and logistics; passenger travel and leisure; media; sports and entertainment; and telecommunications indicate:

Big Data is critical to meet strategic objectives.



Sharing insights is a must-have capability for businesses.



Big Data will amplify other technology innovations.



Nearly **67%** have started to use Big Data in specific cases  
... but only **6%** consider themselves mature.



**62%** view data quality and huge data volumes as major challenges.



**70%** of executives consider improved customer engagement and performance across lines of business as high priorities.



Source: SAP Blogs

**Making Values of data can be challenging, yet we will learn how to achieve this in this QMST5336!!!**

# Analytics: Categorization



We can consider big data issues as a subset of analytics.

The key to analytics is that it provides useful insights and better decision making using available data – either *big* or *small*.

## AmazonGo: Decision Making

**Question:** Whether (& where) should Amazon open physical stores?  
Analytics can help. This is how:

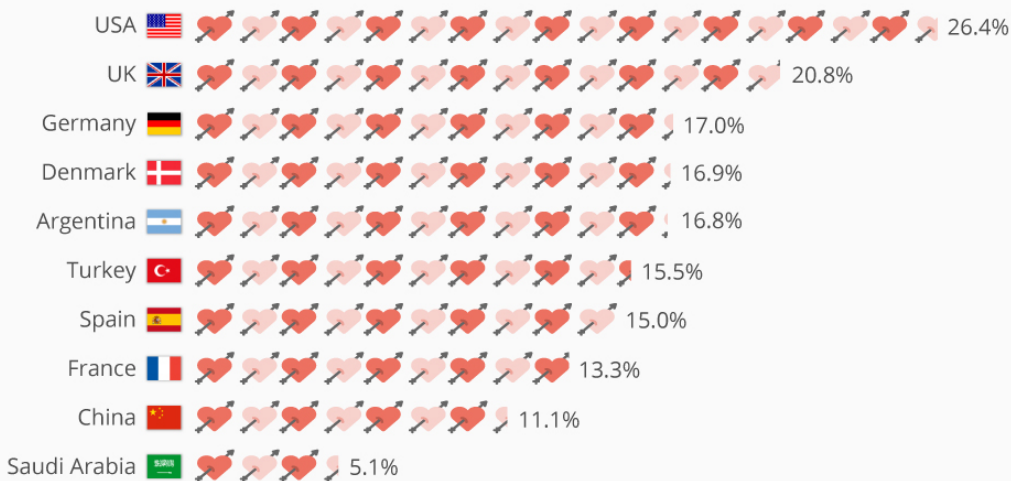
- 1 Descriptive analytics
  - Online shopping data
  - Market analysis
- 2 Predictive analytics
  - Future demand & potential canalization
- 3 Prescriptive analytics
  - Long-term branding, profitability etc.



# Love & Decision Making

## Investing in Love

Share of adult population that are active paying customers of a dating service in 2017



Adults aged 16 and over.

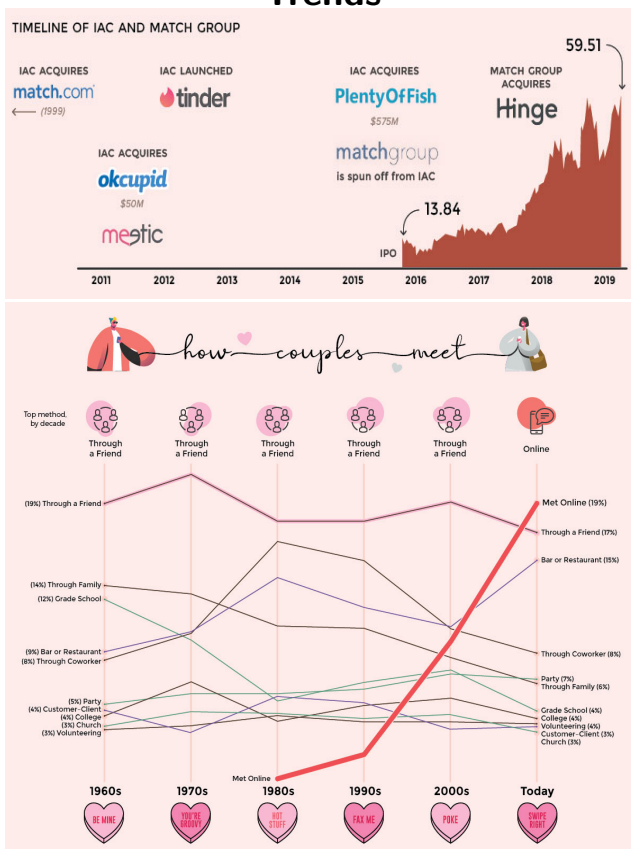
'Dating services' is made up of the following categories:  
Matchmaking, Online Dating, Casual Dating.

CC BY ND  
@StatistaCharts

Source: Statista Digital Market Outlook

statista

## Online Dating Apps Trends



**Question:** How do online dating apps make money?

- 1 Descriptive analytics
  - historical data
  - market analysis
- 2 Predictive analytics
  - classification & matching
- 3 Prescriptive analytics
  - advertising strategies/portfolio
  - pricing subscription fees

# To Do or Not To Do?

We are not debating about online dating:

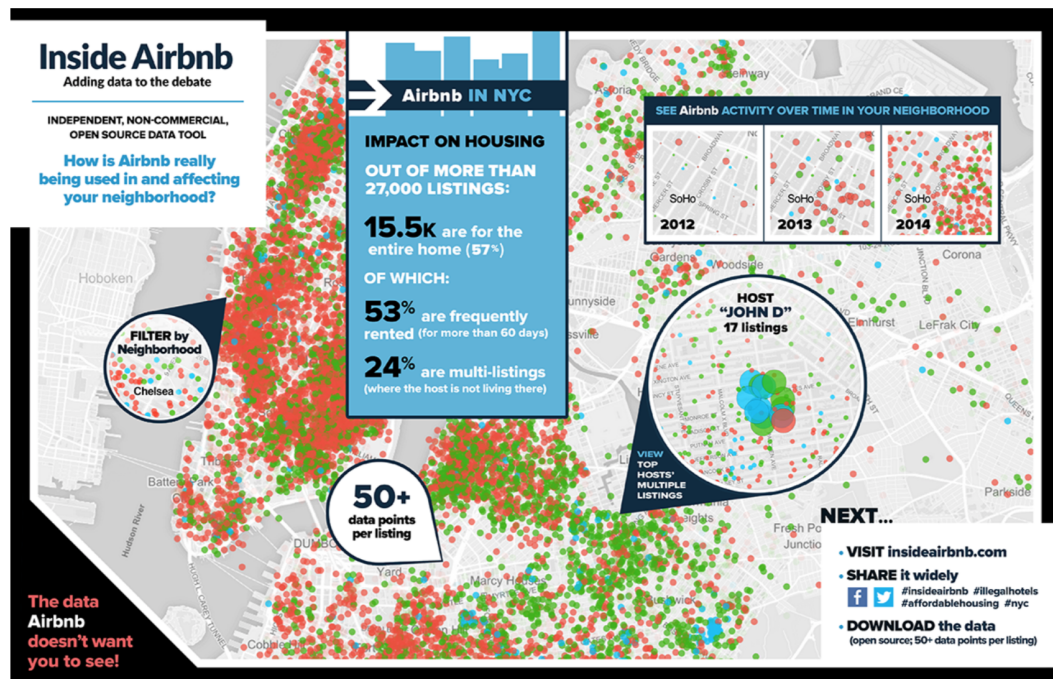
- Every opinion is welcomed (preferably with evidence to back it up)
- We need to use correct analytics to make decisions
- QMST5336 will teach you how to use real-life data to support decision making.



## The Case for this course

We will be mainly using data from **Inside Airbnb**

### About Inside Airbnb



After this course, you are able to decide

**Airbnb** or **rental** business for your investing property in Austin

**Inside Airbnb**  
Adding data to the debate

About Behind Get the Data

Like 4.5K Share Tweet

### Austin, Texas, United States

See [Austin data visually here](#).

| Date Compiled | Country/City | File Name                              | Description                                                                                                     |
|---------------|--------------|----------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| 14 May, 2019  | Austin       | <a href="#">listings.csv.gz</a>        | Detailed Listings data for Austin                                                                               |
| 14 May, 2019  | Austin       | <a href="#">calendar.csv.gz</a>        | Detailed Calendar Data for listings in Austin                                                                   |
| 14 May, 2019  | Austin       | <a href="#">reviews.csv.gz</a>         | Detailed Review Data for listings in Austin                                                                     |
| 14 May, 2019  | Austin       | <a href="#">listings.csv</a>           | Summary information and metrics for listings in Austin (good for visualisations).                               |
| 14 May, 2019  | Austin       | <a href="#">reviews.csv</a>            | Summary Review data and Listing ID (to facilitate time based analytics and visualisations linked to a listing). |
| N/A           | Austin       | <a href="#">neighbourhoods.csv</a>     | Neighbourhood list for geo filter. Sourced from city or open source GIS files.                                  |
| N/A           | Austin       | <a href="#">neighbourhoods.geojson</a> | GeoJSON file of neighbourhoods of the city.                                                                     |

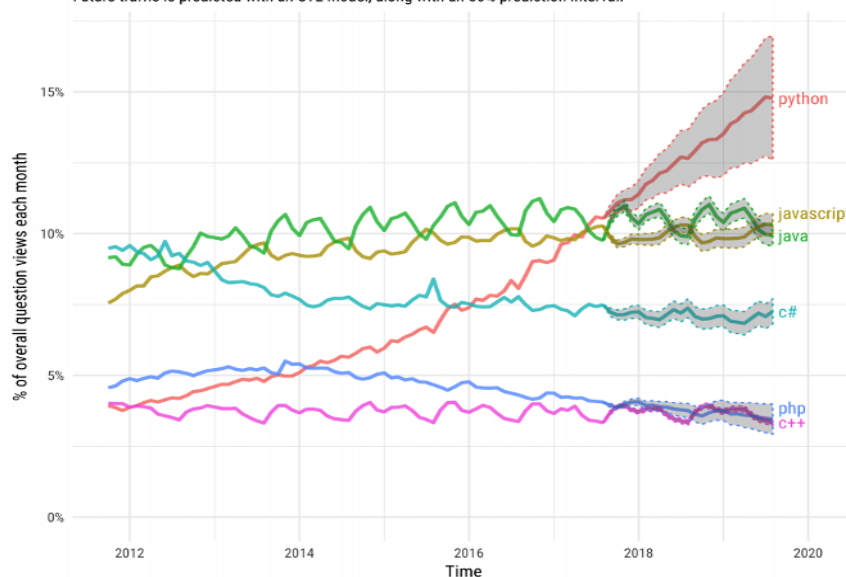
<http://insideairbnb.com/get-the-data.html>

## The Popularity



### Projections of future traffic for major programming languages

Future traffic is predicted with an STL model, along with an 80% prediction interval.



## A Bit Of History



Python's name is derived from the British comedy group *Monty Python*, whom Python creator **Guido van Rossum** enjoyed while developing the language.

Source: Wikipedia

## Why Python



# Python IDEs

## Definition

**Integrated Development Environment (IDE)** is a program dedicated to software development, including coding, execution, debugging and resource management.

**Text editor** can highlight syntax and format codes ONLY.

**Code editor** can execute code and control a debugger besides the function of text editors, but usually smaller, quicker and less feature rich, compared to IDEs.

- General editors and IDEs with Python support
  - Visual Studio, Eclipse+PyDev, Sublime Text & more
- Python-specific editors and IDEs
  - e.g. PyCharm, Spyder, Thonny & more



## Examples of Python IDEs

### IDLE/IdleX



- open source IDE written in Python
- comes with Python installation
- multi-window text editor
- a powerful debugger
- no numbering or multiple-line execution, hard for a project

### PyCharm



- a smart platform with auto code completion, error detection, quick fixing etc
- supports customizable interface
- allows cross-platform development
- community & professional versions
- an expensive tool, the initial installation is difficult

### Jupyter Notebook



- open source web app
- create & maintain notebook documents
- for beginners and educators
- support multiple coding languages, including Python & R
- allows to be shared for a more collaborative touch

# Which Python IDE is Right for YOU?

Some basic recommendations:

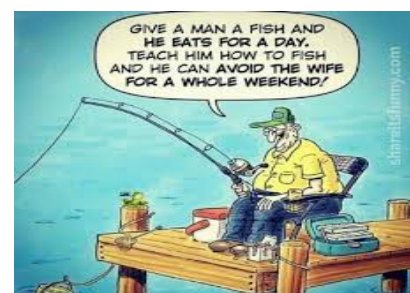
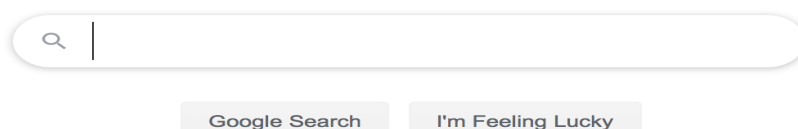
- Almost all these IDEs can work for this course.
- New Python developers should try solutions with as few customizations as possible. The less gets in the way, the better.
- If you're already developing other software, you may find it easier to add Python capabilities to your existing toolset.



## Install Python

**Python 3** can work on

- Windows
- macOS
- Linux
- iOS (iPhone/iPad): *Pythonista app* from the *iOS app store*.
- Android (Phones & Tablets): *Pydroid 3* from the *Google Play store*.
- Online
  - Python.org Online Console: [www.python.org/shell](http://www.python.org/shell)
  - Python Fiddle: [pythonfiddle.com](http://pythonfiddle.com)
  - Repl.it: [repl.it](http://repl.it)
  - Trinket: [trinket.io](http://trinket.io)
  - Python Anywhere: [www.pythonanywhere.com](http://www.pythonanywhere.com)



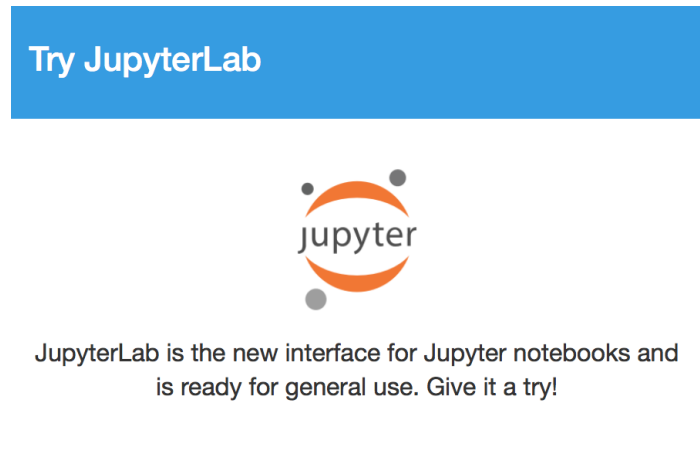
# Jupyter Notebook

You can install Jupyter after Python installation on your computers.

<https://jupyter.org/install>

OR

Try JupyterLab online without installation. <https://jupyter.org/try>



## Install Packages from PyPI

It is common in the Python community to refer to a distribution (i.e. a bundle of software to be installed) using the term “package”.

**Step 1** Ensure you can run pip from the command line (i.e. *Shell, terminal or console*)

```
1 #check if you have pip available
2 pip -- version
3 #If pip isn't already installed, bootstrap it from the standard
  library
4 python -m ensurepip --default-pip
5 # If that still doesn't work, download and run
6 get-pip.py
```

**Step 2** Install packages *SomeDistribution*

```
1 pip3 install --SomeDistribution
```

Upgrade an already installed *SomeDistribution* to the latest

```
1 pip3 install --upgrade SomeDistribution
```

<https://packaging.python.org/tutorials/installing-packages/>

## Before Next Lecture

- Install Python on your PC  
**OR**
- Find your online IDE for Python
- Install Python packages
  - Numpy, Pandas, SciPy, scikit-learn, Statsmodels



Make sure to complete “Introduce ME” on Canvas if you have NOT done so.